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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	No Idea
Selected Domain	Domain 1 — Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 8 — Decent Work and Economic Growth (with SDG 4 as supporting impact)
Project / Solution Name	NexPath AI
Name (Leader)	Gan Yu Xuan
Email & Phone (Leader)	yuxuan0118.yx@gmail.com 013-404 5395
University Affiliation(s)	Universiti Teknologi Malaysia, Universiti Malaya
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

In this fast-paced world, AI is rapidly changing job responsibilities and the skills employers expect. However, students and fresh graduates often struggle to identify growing career opportunities, necessary skills, and ways to prepare for changing workforce demands.

Domain 1 highlights Malaysia's skills-development challenge: 56% of professionals use AI tools at work, while only 26% have received formal AI training, according to the Hays Asia Salary Guide 2025. The Ministry of Science, Technology and Innovation also estimates that over 600,000 workers will need reskilling within three to five years.

Our initial target users are Computer Science students and fresh graduates who have programming knowledge from coursework and projects but lack clear, evidence-based career guidance. Existing resources often provide job listings or generic learning recommendations without connecting current skills, labour-market trends, future job requirements, and actionable learning steps.

Resume reliability is another challenge. Students may struggle to tailor resumes to job descriptions, while generative AI may produce unsupported skills, responsibilities, or achievements.

NexPath AI addresses these challenges through skills profiling, sourced AI and labour-market trends, explainable career matching, personalised reskilling pathways, and SDG mapping. Its evidence-based resume builder will reorganise and rephrase only user-confirmed information, helping students present their capabilities without fabricating experience.

The project supports SDG 8 by improving career readiness and SDG 4 by making skills development more accessible. Ultimately, NexPath AI aims to help users make informed career decisions while keeping human judgement and user approval central.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

NexPath AI is an AI-assisted career navigation platform initially designed for Computer Science students and fresh graduates. Users upload a résumé or enter their education, skills, projects and interests. The system extracts candidate information into editable fields; nothing extracted is treated as verified until the user reviews and confirms it.

A career exploration engine compares confirmed capabilities with structured role-skill profiles and explains recommended roles, matching strengths and missing competencies. A visual Future Skills Radar presents skill demand and readiness through interactive charts, rather than lengthy AI-generated text. The personalised roadmap prioritises learning actions, portfolio projects and measurable milestones. Career Portfolio connects selected roles and skills to relevant SDGs, while an evidence-based résumé builder improves wording without inventing credentials or achievements.

The prototype uses a responsive web interface, a structured data layer for user profiles and career-skill mappings, and an AI inference service for extraction, explanation and learning suggestions. Proposed Phase 1 workshop applications include GenAI API integration, database architecture and cloud deployment. Replit is part of the team's development workflow; RunPod is the intended model-inference integration, subject to confirmation of its live configuration. AI outputs are constrained by user-confirmed information, displayed with concise explanations and subject to user correction.

By Phase 2, the team aims to deploy an end-to-end demonstrator covering profile creation, explainable matching, visual skills gaps and an actionable roadmap, with responsible handling of résumé data.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	Planned approach / verification
Frontend	React + Vite web prototype; interactive charts
Backend	API/ backend layer
AI inference	RunPod inference endpoint ; model-assisted résumé parsing and explanations.
Data	Structured career/skill catalogue and user-confirmed profile
Developement and hosting	VS Code, GitHub Copilot, Replit
Sponsor resource	RunPod and Replit credits were discussed as hackathon resources

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We begin with a defined user group: Computer Science students and fresh graduates who need actionable career guidance. The team frames the problem around three tasks: understand existing capabilities, identify suitable future roles and decide what to learn next. The prototype therefore prioritises a short onboarding flow, visual comparisons and actionable cards over long paragraphs.

Development follows an iterative build-test-refine cycle. First, map representative user journeys and create sample profiles and role-skill records. Next, implement résumé ingestion with an explicit correction screen, career matching, interactive radar visualisation and a prioritised learning roadmap. Then, conduct task-based usability checks using test profiles to measure completion, clarity and extraction correctness. Feedback will guide revisions to labels, chart readability, mobile layout and AI explanations.

The Phase 2 deployment target is a browser-accessible demonstration with documented fallback behaviour when AI inference is unavailable. User confirmation is mandatory before résumé facts are reused. The team will verify API credentials, privacy controls, repository access and reproducibility before final judging.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.5: full and productive employment and decent work; supporting SDG 4.4: increase relevant technical and vocational skills.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Published contextual baseline cited in draft: 26% formal AI training among surveyed professionals. Product-specific baseline:
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage of pilot users who can identify a suitable role, correctly interpret their skills gap, and produce a three-step learning plan without assistance; résumé extraction field accuracy.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed pilot targets (not results): ≥80% task completion across ≥10 test users; ≥90% correct extraction of explicitly stated résumé fields on a labelled test set; ≥80% users identify three relevant next steps.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	code	MIT
Vite	code	MIT
Express	code	MIT
sentence-transformers	code	Apache-2.0
all-MiniLM-L6-v2	model	Apache-2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Gan Yu Xuan

[24 October 2026]